

Telecommunication Assignment -6



Faculty of Engineering

Department of Electrical Engineering and Computer Science

Hochschule Wismar University of Applied Sciences

Technology, Business and Design

Telecommunication Project 2023

Guided by : Prof. Dr.-Ing Toralf Renkwitz

Date : 01-06-2023

Anusree Kundu	(460596)
Chandrayudula Ramachandra Srinivasa Prasad Tharun Kumar	(433303)
Lohith Reddy Prane	(431031)
Neha Manjunath	(451772)
Shreyas Ravi Vasista	(457183)

LIST OF CONTENTS

Serial No	Title	Page No
1	Task - 1	1
2	Task - 2	4
3	Task - 3	5
4	References	7

LIST OF FIGURES

Fig No	Title	Page No
1.1	Radiation Pattern at 0.5 Wl	2
1.2	Cross-Section at 0.5 Wl	2
1.3	Radiation Pattern at 1.0 Wl	2
1.4	Cross-Section at 1.0 Wl	2
1.5	Radiation Pattern at 1.5 Wl	3
1.6	Cross-Section at 1.5 Wl	3
2.1	Array geometry at 1.0 Wl	4
2.2	Array geometry at 1.5 Wl	4
3.1	Radiation Pattern(Equ) at 0.5 Wl	6
3.2	Cross-Section(Equ) at 0.5 Wl	6
3.3	Radiation Pattern(Equ) at 1.0 Wl	6
3.4	Cross-Section(Equ) at 1.0 Wl	6
3.5	Radiation Pattern(Equ) at 1.5 Wl	6
3.6	Cross-Section(Equ) at 1.5 Wl	6

Task - 1

Simulate the radiation pattern (Array factor) of different antenna array arrangements. 5 x 5 antenna , squared grid, 0.5 lambda wavelength (WL) spacing for vertical and 60° off-vertical beam pointing along one array axis. Plot the resulting radiation pattern in spherical top view (directional cosines) and one cross section through the main beam. Repeat same for 1.0 and 1.5 WL

Script :

```
import numpy as np
import matplotlib.pyplot as plt

def radiation_pattern(theta, phi, n, d, wl):
    k = 2 * np.pi / wl # Wave number
    af = np.zeros_like(theta, dtype=complex) # Initialize array factor with complex dtype

    for i in range(n):
        for j in range(n):
            psi = k * d * (i * np.sin(phi) * np.cos(theta) + j * np.sin(phi) * np.sin(theta))
            af += np.exp(1j * psi)
    af /= (n ** 2) # Normalize array factor
    return np.abs(af) ** 2

# Parameters
n = 5 # Number of elements in each row/column
array_size = 1.0 # Spacing between elements (in wavelengths)
wl = 0.5 # Change this value for 1.0 and 1.5 for other cases

# Theta and phi angles
theta = np.linspace(0, 2 * np.pi, 360)
phi = np.radians(60)

# Calculate array factor (radiation pattern)
pattern = radiation_pattern(theta, phi, n, array_size, wl)

# Plot radiation pattern in spherical top view
```

```
plt.polar(theta, pattern)
plt.title("Radiation Pattern (Spherical Top View)")
plt.ylim(0, np.max(pattern))
plt.show()

# Plot cross-section through the main beam
plt.plot(np.degrees(theta), pattern)
plt.title("Cross-Section through the Main Beam")
plt.xlabel("Theta (degrees)")
plt.ylabel("Normalized Power")
plt.xlim(0, 360)
plt.ylim(0, np.max(pattern))
plt.grid(True)
plt.show()
```

Radiation Pattern (Spherical Top View) with Equilateral grid structure

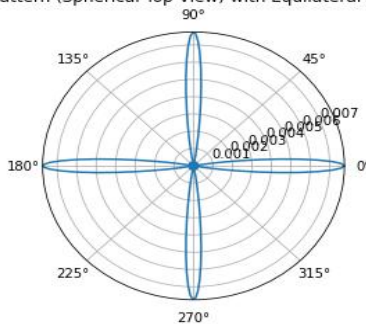


Fig 1.1 : Radiation Pattern at 0.5 Wl

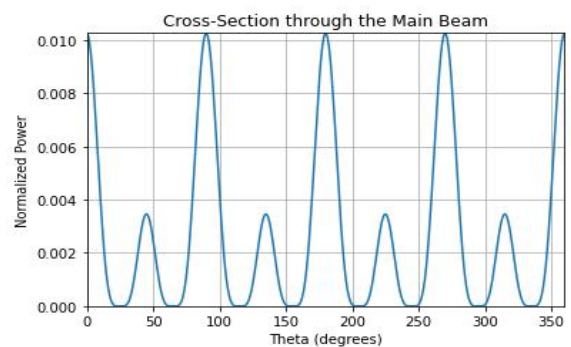


Fig 1.2 : Cross-Section at 0.5 Wl

Radiation Pattern (Spherical Top View)

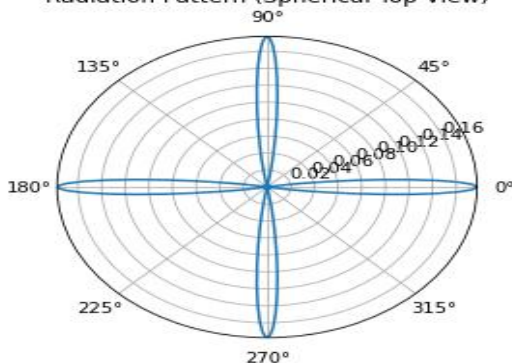


Fig 1.3 : Radiation Pattern at 1.0 Wl

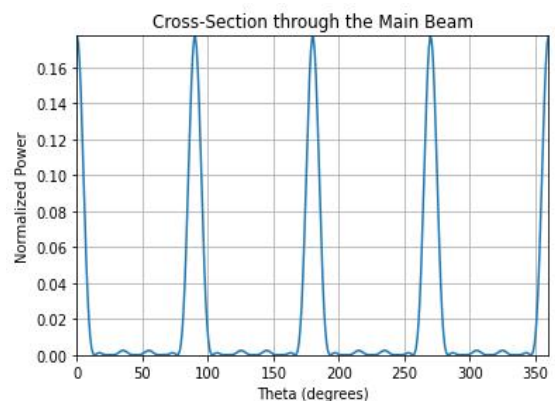


Fig 1.4 : Cross-Section at 1.0 Wl

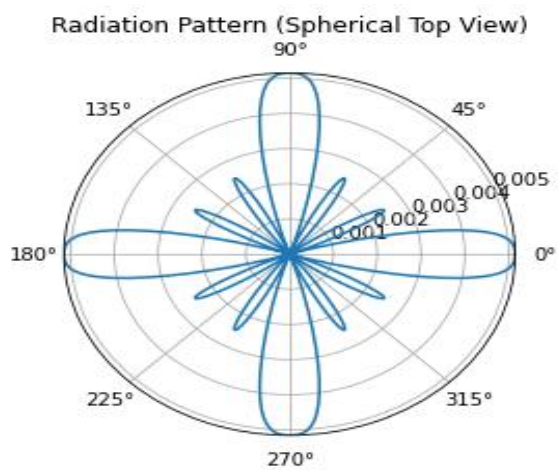


Fig 1.5 : Radiation Pattern at 1.5 Wl

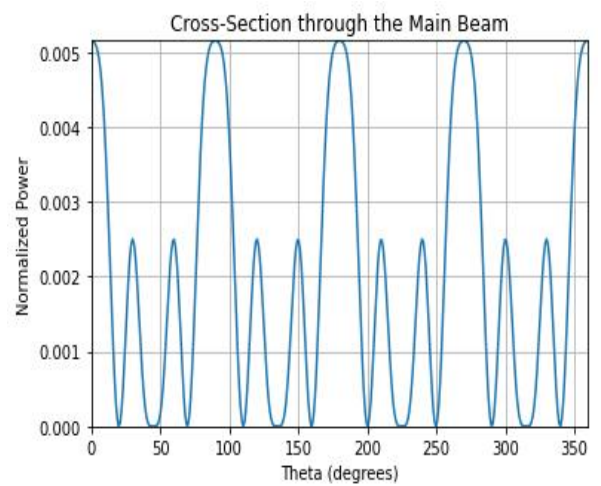


Fig 1.6 : Cross-Section at 1.5 Wl

Task - 2

Plot the array geometry and mark the phase distribution for the off-vertical scenario for 0.5 WL and 1.5 WL.

Script :

```
def plot_array_geometry(array_spacing, n):  
    plt.figure()  
    plt.scatter(array_spacing * np.arange(n), np.zeros(n), marker='o')  
    plt.title('Array Geometry')  
    plt.xlabel('X-axis')  
    plt.ylabel('Y-axis')  
    plt.grid(True)  
  
# Plot array Geometry and mark the phase distribution  
array_spacing = 0.5 * wl  
plot_array_geometry(array_spacing, n)  
plt.title('Array Geometry')  
plt.xlabel('X-axis (m)')  
plt.ylabel('Y-axis (m)')
```

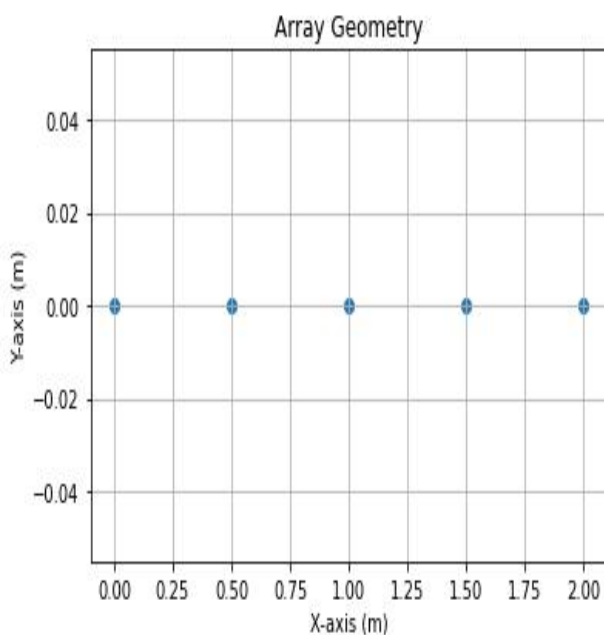


Fig 2.1 : Array geometry at 1.0 WL

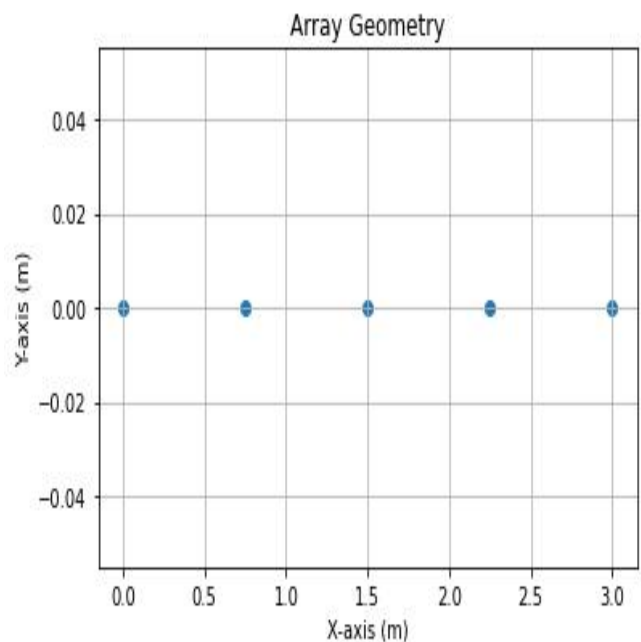


Fig 2.2 : Array geometry at 1.5 WL

Task - 3

Same as task -1 but for equilateral grid structure

Script :

```
def radiation_pattern_Equ(theta, phi, n, d, wl):
    k = 2 * np.pi / wl # Wave number
    af = np.zeros_like(theta, dtype=complex) # Initialize array factor with complex dtype
    for i in range(n_1):
        for j in range(n_1):
            psi = k * d * (i * np.sin(phi) * np.cos(theta) + j * np.sin(phi + np.pi/3) * np.sin(theta))
            af += np.exp(1j * psi)
    af /= (n * (n + 1) // 2)

# Plot radiation pattern in spherical top view (Equilateral grid structure)
n_1 = 10
pattern_1 = radiation_pattern(theta, phi, n_1, array_size, wl)
plt.polar(theta, pattern_1)
plt.title("Radiation Pattern (Spherical Top View) with Equilateral grid structure")
plt.ylim(0, np.max(pattern_1))
plt.show()

# Plot cross-section through the main beam (Equilateral grid structure)
plt.plot(np.degrees(theta), pattern_1)
plt.title("Cross-Section through the Main Beam with Equilateral grid structure")
plt.xlabel("Theta (degrees)")
plt.ylabel("Normalized Power")
plt.xlim(0, 360)
plt.ylim(0, np.max(pattern_1))
plt.grid(True)
plt.show()
```

Radiation Pattern (Spherical Top View) with Equilateral grid structure

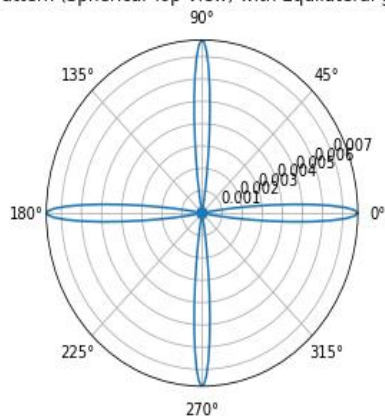


Fig 3.1 : Radiation Pattern(Equ) at 0.5 Wl

Cross-Section through the Main Beam with Equilateral grid structure

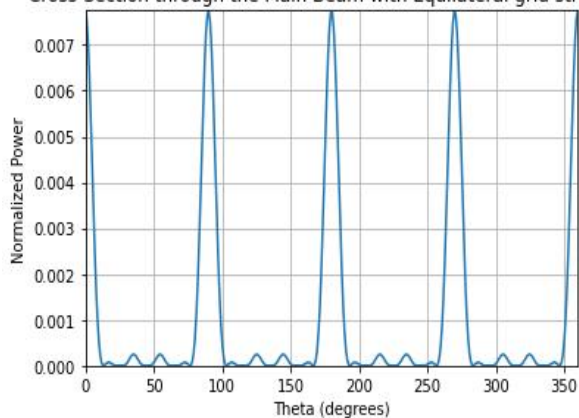


Fig 3.2 : Cross-Section(Equ) at 0.5 Wl

Radiation Pattern (Spherical Top View) with Equilateral grid structure

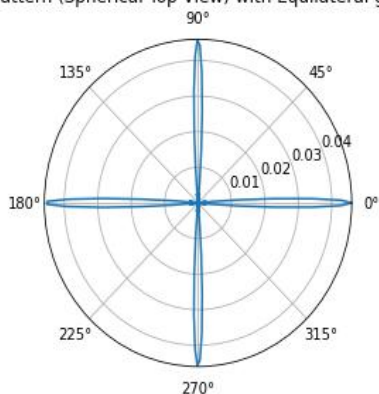


Fig 3.3 : Radiation Pattern(Equ) at 1.0 Wl

Cross-Section through the Main Beam with Equilateral grid structure

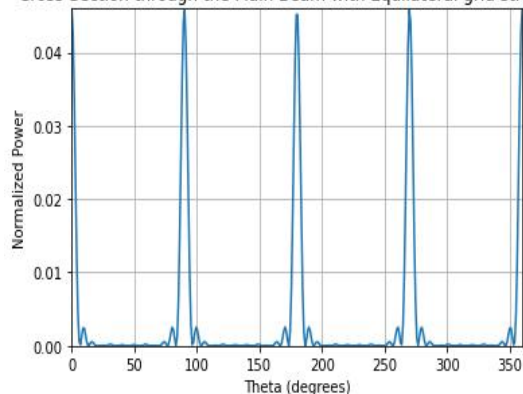


Fig 3.4 : Cross-Section(Equ) at 1.0 Wl

Radiation Pattern (Spherical Top View) with Equilateral grid structure

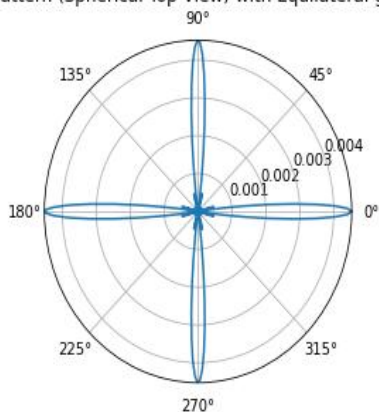


Fig 3.5 : Radiation Pattern(Equ) at 1.5 Wl

Cross-Section through the Main Beam with Equilateral grid structure

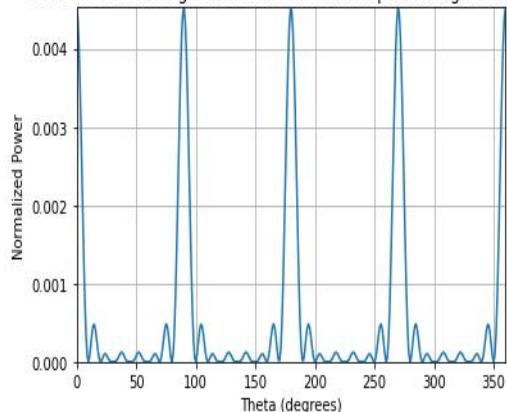


Fig 3.6 : Cross-Section(Equ) at 1.5 Wl

References

- [1] John Grant 'Antenna Arrays and Python - The Array ' [online]
<https://medium.com/python-pandemonium/antenna-arrays-and-python-the-array-finally-3613966153d7> [accessed 30.05.2023]
- [2] 'Modelling an antenna array' [online]
<https://scipython.com/book2/chapter-7-matplotlib/examples/modelling-an-antenna-array/> [accessed 31.05.2023]
- [3] Prof. Sean Victor Hum 'Radio and Microwave Wireless system' [online]
<https://www.waves.utoronto.ca/prof/svhum/ece422/notes/15-arrays2.pdf>
[accessed 31.05.2023]